

Taiwan

Taiwan Fair Trade Commission

The booming digital economy has brought both innovation and challenges to markets. In 2023, the Taiwan Fair Trade Commission (hereinafter ‘the TFTC’) finalized and published its *White Paper on Competition Policy in the Digital Economy* (hereinafter ‘the White Paper’). The White Paper explores competition issues that may be triggered by tech giants from the perspective of four aspects of the digital economy, namely, ‘Platforms are intermediaries for transactions,’ ‘Data is a contested resource,’ ‘Market expansion is the path to growth,’ and ‘Concentration of market power is the trend in digital markets.’ The White Paper also outlines possible enforcement positions and tendencies of the TFTC in the future. As for the adoption and application of computational tools to improve enforcement efficiency, the TFTC has already begun to strengthen its ability to engage in market monitoring and case analysis with such tools, thereby laying the groundwork for competition assessments in the digital era. The accomplishments of the TFTC regarding the application of computational tools are summarized below and are followed by a discussion on the challenges it has faced during their enforcement.

I. Accomplishments in Applying Computational Tools

The TFTC utilizes data-mining techniques and data-visualization tools through open-source applications such as R and Python to facilitate the sorting and analysis of case information.

1. In the cinema merger case and the food stall owners’ price-fixing case, the relevant geographic markets were defined using isochrone (catchment area) mapping. Relevant computational tools were applied to estimate and map the accessible area within a certain distance or travel time by vehicle, motorcycle, or on foot. With the R programming language, the isochrone ranges of travel time were converted into visualized spatial areas, facilitating the analysis of the scope of the geographic market for each case. Compared to the limitations of the traditional approach to defining the geographic markets in terms of administrative zoning, such an analytical approach enables a more straightforward calculation of market shares for the relevant enterprises in the geographic market.
2. In the carbonated beverages price-increased case, an interrupted time series model was established with computational tools to process the price/volume

big data provided by enterprises, taking the monthly effect into consideration. Regression models were built for various carbonated beverages to estimate product demand elasticities by analyzing differences before and after price adjustments. The economic reasonableness of the price adjustments made by the enterprises was assessed by comparing demand elasticities and the magnitude of price adjustments.

The TFTC engages in the monitoring of multiple essential commodities such as wheat, flour, and soybeans, applying web scraping tools to continuously monitor price fluctuations of major imported commodities and to build a continuously updated dataset. When anomalies are detected, investigations are initiated by the responsible departments along with analysis employing quantitative methodologies such as Q-learning and cointegration analysis.

The TFTC also analyzes anti-competitive effects arising from mergers using logistic regression models. The traditional GUPPI fails to translate upward pricing pressure into a ratio of price increases. By contrast, the TFTC has built logistic regression models to assess competition dynamics using the market shares of the merging parties, as well as product characteristics and price data. The ratios of post-merger price increases can be estimated, which helps address the limitations of the GUPPI analysis.

In addition to its own initiatives and the application of computational tools as mentioned above, the TFTC also commissions external scholars or experts on an ad hoc basis to offer relevant research reports for its reference. Examples of research topics include ‘Research on Major Anti-competition Issues of Algorithms and Cartels,’ ‘Research on the Empirical Analysis of Collusive Behaviors in Pricing Strategies in the Digital Economy,’ ‘Assisting the Introduction of Algorithmic Detection Mechanisms and Empirical Studies for Market Definition,’ ‘Applying Big Data and Machine Learning Methods for Cartel Detection,’ and ‘Applying Big Data and Algorithms for Cartel Detection.’

II. Enforcement Challenges and Future Plans

In confronting the challenges posed by competition in the era of the digital economy, the TFTC has proactively adopted computational tools to enhance the effectiveness of enforcement, achieving tangible results in practical application. For instance, by leveraging open-source programming languages such as R and Python, combined with data mining and visualization techniques, the TFTC has applied

computational tools to conduct economic analysis in relevant cases, significantly improving the efficiency and quality of case analysis.

Regarding the use of computational tools to support competition law enforcement, the TFTC has not yet established a dedicated unit or temporary task force. Consequently, due to a shortage of relevant professional expertise, the TFTC typically commissions external scholars or experts to conduct studies involving advanced applications of computational tools and to offer research reports for enforcement reference.

Looking ahead, the TFTC plans to incorporate Artificial Intelligence (AI) into its existing competition law case database search system. Specifically, in line with AI developments, the TFTC intends to integrate its historical case databases with generative AI technologies, thereby enabling advanced applications such as case indexing, searching, and summarization.